What Solana Teaches Us About the Future of Scalable Blockchains

Solana as a case study: real load, TPS vs usability, and requirements for next-gen infrastructure

Solana is not a benchmark

> A production stress test for the entire idea of high-throughput L1.

- users and wallets
- DEXes, arbitrage, MEV, memecoins
- RPC, validators, priority fees
- outages, postmortems, upgrades

The thesis

Future scalability is measured not by peak TPS, but by the reliability of a user action under load.

Question: can a user complete an important transaction when everyone else is trying to do the same?

Usability metrics

Inclusion latency
time to inclusion in a block

Drop rate
share of transactions lost before confirmation

Fee predictability
accuracy of priority fee estimation

RPC availability
endpoint stability

What real load looks like

Not "many transactions" in general.

It is when thousands of actors compete for the same outcome:

- be first into a token launch
- arbitrage DEX pools
- close a position on a price move
- send a swap before state changes

Bottleneck moves

Consensus
forks, votes

Networking
packet loss, QUIC

Scheduler
account contention

Fee market
priority fees

RPC
overload, errors

Wallet UX
retries, status

Low fees are a feature and an attack surface

If transactions are cheap, spam is cheap too.

Case: April 30, 2022 (`04-30-22-solana-mainnet-beta-outage-report-mitigation`):
NFT mint bots, ~6M transaction requests/sec.

So the question is not "how to keep it nearly free always."

The question is: **how to serve normal load cheaply and fairly allocate scarcity at peaks**.

Local fee markets

> Payments shouldn't get more expensive just because one DEX pool got hot.

Idea: price contested state, not the whole network.

Ref: Umbra Research, "Toward Multidimensional Solana Fees"

A local fee market tries to price concrete scarcity:

- accounts
- programs
- compute
- execution queues

RPC is part of scalability

When RPC degrades, the user thinks the blockchain degraded.

Case: Jan 8, 2023: Public RPC outage; block production was not impacted.

Study: Anza, "Transaction Landing on TPU": latency + drop rate.

A production app must be able to:

- fetch a fresh blockhash
- track expiration via lastValidBlockHeight
- retry safely and rebuild the transaction

Reliability > peak numbers

February 2024: Mainnet Beta halt of about 5 hours (`02-06-24-solana-mainnet-beta-outage-report`).

Official postmortem: bug in `LoadedPrograms`, legacy loader, and an infinite recompilation loop.

Lesson: a high-performance blockchain is a complex software system.

Next-gen infrastructure requirements

- user-perceived reliability over vanity TPS
- local and transparent fee markets
- adaptive clients for normal/congested/degraded modes
- production-grade RPC and retries
- observability on inclusion, fees, latency, drops
- multi-client resilience

Where Solana is moving

Already on mainnet:

- XDP — faster packet processing (Agave 3.0.9+)

Roadmap, target Agave 4.1:

- Alpenglow: consensus with ~150 ms confirmation
- 100M compute units per block: more capacity
- larger transactions: more composability

In parallel: Firedancer — a separate validator client (Jump Crypto).

Closing takeaway

> The future of scalable blockchains is not maximum TPS. It is the ability to stay useful at the worst moment.

The new metric: can users reliably get important transactions included under real load?

Slides and sources are on GitHub

github.com/kalaninja/solana-scalability-lessons

Thank you.